

Study Notes: Determinants

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1 Determinants

1.1 Definition and Notation

Let A be an $n \times n$ (square) matrix with real entries. The *determinant* of A is a scalar, denoted $\det(A)$ or $|A|$, that encodes important algebraic and geometric information about the linear transformation represented by A .

Key viewpoints:

- Algebraically, $\det(A) \neq 0$ if and only if A is invertible.
- Geometrically, for a linear map $T_A : \mathbb{R}^n \rightarrow \mathbb{R}^n$ given by $T_A(\mathbf{x}) = A\mathbf{x}$, the absolute value $|\det(A)|$ is the factor by which A scales n -dimensional volume.
- The sign of $\det(A)$ indicates whether orientation is preserved ($\det(A) > 0$) or reversed ($\det(A) < 0$).

1.2 Small-Size Determinants

2×2 case. For

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix},$$

the determinant is

$$\det(A) = ad - bc.$$

3×3 case via cofactor expansion. For

$$A = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix},$$

a common formula (expansion along the first row) is

$$\det(A) = a_{11} \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} - a_{12} \begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix} + a_{13} \begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix},$$

where each 2×2 determinant is computed as in the 2×2 case.

1.3 Minors, Cofactors, and Cofactor Expansion

Let $A = (a_{ij})$ be an $n \times n$ matrix.

- The *minor* M_{ij} of a_{ij} is the determinant of the $(n-1) \times (n-1)$ matrix obtained by deleting row i and column j from A .
- The *cofactor* C_{ij} of a_{ij} is

$$C_{ij} = (-1)^{i+j} M_{ij}.$$

The determinant of A can be computed by *cofactor expansion* along any row i :

$$\det(A) = \sum_{j=1}^n a_{ij} C_{ij},$$

or along any column j :

$$\det(A) = \sum_{i=1}^n a_{ij} C_{ij}.$$

This formula is known as *Laplace expansion*. A practical strategy is to expand along a row or column that contains many zeros to minimize computation.

1.4 Row Operations and Determinant

Elementary row operations affect the determinant in specific ways:

1. Swapping two rows multiplies the determinant by -1 .
2. Multiplying a row by a nonzero scalar k multiplies the determinant by k .
3. Adding a multiple of one row to another row does *not* change the determinant.

These properties allow efficient computation of determinants by row-reducing a matrix to an upper triangular form and tracking how the determinant changes.

2 Identity Matrices

2.1 Definition

The $n \times n$ *identity matrix* I_n is the matrix with ones on the main diagonal and zeros elsewhere:

$$I_n = \begin{pmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{pmatrix}.$$

2.2 Key Properties

- I_n is the multiplicative identity for matrix multiplication:

$$I_n A = A I_n = A$$

for any $n \times n$ matrix A .

- The determinant of the identity matrix is 1:

$$\det(I_n) = 1.$$

- The identity matrix represents the linear transformation that leaves every vector unchanged.

3 Inverse Matrices (Cofactor/Adjoint Method)

3.1 Definition

An $n \times n$ matrix A is called *invertible* (or *nonsingular*) if there exists a matrix A^{-1} such that

$$A A^{-1} = A^{-1} A = I_n.$$

If no such matrix exists, A is called *singular*.

3.2 Adjugate (Adjoint) Matrix

The *adjugate* (or *adjoint*) of A , denoted $\text{adj}(A)$, is defined as

$$\text{adj}(A) = C^T,$$

where C is the matrix of cofactors:

$$C = (C_{ij}), \quad C_{ij} = (-1)^{i+j} M_{ij}.$$

A fundamental identity relating A , its adjugate, and the identity matrix is

$$A \cdot \text{adj}(A) = \text{adj}(A) \cdot A = \det(A) \cdot I_n.$$

3.3 Inverse via Adjugate

If $\det(A) \neq 0$, then A is invertible and the inverse can be expressed using the adjugate:

$$A^{-1} = \frac{1}{\det(A)} \text{adj}(A).$$

This formula is exact but becomes computationally expensive for large n (because it requires many cofactors). For small matrices (e.g. 2×2 or 3×3) it is often reasonable.

3.4 Example (2×2)

Let

$$A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}.$$

First compute the determinant:

$$\det(A) = 1 \cdot 4 - 2 \cdot 3 = -2.$$

Cofactors:

$$C_{11} = 4, \quad C_{12} = -3, \quad C_{21} = -2, \quad C_{22} = 1.$$

Cofactor matrix:

$$C = \begin{pmatrix} 4 & -3 \\ -2 & 1 \end{pmatrix}.$$

Adjugate:

$$\text{adj}(A) = C^T = \begin{pmatrix} 4 & -2 \\ -3 & 1 \end{pmatrix}.$$

Inverse:

$$A^{-1} = \frac{1}{\det(A)} \text{adj}(A) = \frac{1}{-2} \begin{pmatrix} 4 & -2 \\ -3 & 1 \end{pmatrix} = \begin{pmatrix} -2 & 1 \\ \frac{3}{2} & -\frac{1}{2} \end{pmatrix}.$$

4 Transpose Matrices

4.1 Definition

Given an $m \times n$ matrix $A = (a_{ij})$, its *transpose* A^T is the $n \times m$ matrix obtained by interchanging rows and columns:

$$(A^T)_{ij} = a_{ji}.$$

In words, the i -th row of A becomes the i -th column of A^T (and vice versa).

4.2 Basic Properties

For matrices A and B of compatible sizes, and scalars α, β :

- $(A^T)^T = A$.
- $(A + B)^T = A^T + B^T$.
- $(\alpha A)^T = \alpha A^T$.
- $(AB)^T = B^T A^T$.

If A is $n \times n$, then so is A^T . In that case, the determinant satisfies

$$\det(A^T) = \det(A).$$

This means that transposing a square matrix does not change its determinant.

5 Singular Matrices and Unique Solutions

5.1 Singular vs Nonsingular

For an $n \times n$ matrix A :

- A is *nonsingular* (invertible) if $\det(A) \neq 0$.
- A is *singular* (non-invertible) if $\det(A) = 0$.

5.2 Relationship to Solutions of $A\mathbf{x} = \mathbf{b}$

Consider the linear system

$$A\mathbf{x} = \mathbf{b},$$

where A is $n \times n$.

- If $\det(A) \neq 0$ (nonsingular A), then:
 - A has an inverse A^{-1} .
 - The system has a *unique* solution:

$$\mathbf{x} = A^{-1}\mathbf{b}.$$

- If $\det(A) = 0$ (singular A), then:
 - A does not have an inverse.
 - The system either has no solution or infinitely many solutions.
 - There is *never* a unique solution in this case.

Thus, for square systems, the determinant acts as a test for the existence of a unique solution.

6 Cramer's Rule

6.1 Statement

Cramer's rule provides a formula for the unique solution of a square linear system using determinants.

Let A be an $n \times n$ matrix with $\det(A) \neq 0$, and consider the system

$$A\mathbf{x} = \mathbf{b}, \quad \mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{pmatrix}.$$

For each $i \in \{1, \dots, n\}$, define the matrix A_i by replacing the i -th column of A with the vector $\mathbf{b}$:

$$A_i = (\text{col}_1(A) \quad \cdots \quad \text{col}_{i-1}(A) \quad \mathbf{b} \quad \text{col}_{i+1}(A) \quad \cdots \quad \text{col}_n(A)).$$

Then Cramer's rule states:

$$x_i = \frac{\det(A_i)}{\det(A)}.$$

6.2 Conditions and Interpretation

- The formula is valid only if $\det(A) \neq 0$ (so A is invertible and the solution is unique).
- Each component x_i is expressed as a ratio of determinants, showing directly how the right-hand side $\mathbf{b}$ affects each variable.
- Although conceptually elegant, Cramer's rule is inefficient for large systems because computing many determinants is expensive. It is mainly used for small systems or theoretical arguments.

6.3 Example: 2×2 System

Solve

$$\begin{cases} x + 2y = 5, \\ 3x + 4y = 6. \end{cases}$$

Coefficient matrix and vector:

$$A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} 5 \\ 6 \end{pmatrix}.$$

Compute the determinant of A :

$$\det(A) = 1 \cdot 4 - 2 \cdot 3 = -2 \neq 0,$$

so there is a unique solution.

Compute $x_1 = x$. Replace the first column by $\mathbf{b}$:

$$A_1 = \begin{pmatrix} 5 & 2 \\ 6 & 4 \end{pmatrix}.$$

Then

$$\det(A_1) = 5 \cdot 4 - 2 \cdot 6 = 20 - 12 = 8.$$

So

$$x = x_1 = \frac{\det(A_1)}{\det(A)} = \frac{8}{-2} = -4.$$

Compute $x_2 = y$. Replace the second column by $\mathbf{b}$:

$$A_2 = \begin{pmatrix} 1 & 5 \\ 3 & 6 \end{pmatrix}.$$

Then

$$\det(A_2) = 1 \cdot 6 - 5 \cdot 3 = 6 - 15 = -9.$$

So

$$y = x_2 = \frac{\det(A_2)}{\det(A)} = \frac{-9}{-2} = \frac{9}{2}.$$

Thus the unique solution is

$$(x, y) = \left(-4, \frac{9}{2} \right).$$

Determinant of a 2×2 matrix

Consider

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}.$$

The determinant of A is

$$\det(A) = \begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc.$$

Determinant of a 3×3 matrix

Let

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}.$$

A cofactor expansion along the first row gives

$$\det(A) = a_{11} \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} - a_{12} \begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix} + a_{13} \begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix}.$$

Using the 2×2 formula inside, this becomes

$$\det(A) = a_{11}(a_{22}a_{33} - a_{23}a_{32}) - a_{12}(a_{21}a_{33} - a_{23}a_{31}) + a_{13}(a_{21}a_{32} - a_{22}a_{31}).$$

Equivalently (Sarrus' rule):

$$\det(A) = a_{11}a_{22}a_{33} + a_{12}a_{23}a_{31} + a_{13}a_{21}a_{32} - a_{13}a_{22}a_{31} - a_{11}a_{23}a_{32} - a_{12}a_{21}a_{33}.$$

Determinant of a 4×4 matrix

Let

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \\ a_{41} & a_{42} & a_{43} & a_{44} \end{bmatrix}.$$

The cofactor C_{1j} of the entry a_{1j} is

$$C_{1j} = (-1)^{1+j} M_{1j},$$

where M_{1j} is the determinant of the 3×3 minor obtained by deleting row 1 and column j .

A cofactor expansion along the first row is

$$\det(A) = a_{11}C_{11} + a_{12}C_{12} + a_{13}C_{13} + a_{14}C_{14}.$$

Explicitly,

$$\det(A) = a_{11} \begin{vmatrix} a_{22} & a_{23} & a_{24} \\ a_{32} & a_{33} & a_{34} \\ a_{42} & a_{43} & a_{44} \end{vmatrix} - a_{12} \begin{vmatrix} a_{21} & a_{23} & a_{24} \\ a_{31} & a_{33} & a_{34} \\ a_{41} & a_{43} & a_{44} \end{vmatrix} + a_{13} \begin{vmatrix} a_{21} & a_{22} & a_{24} \\ a_{31} & a_{32} & a_{34} \\ a_{41} & a_{42} & a_{44} \end{vmatrix} - a_{14} \begin{vmatrix} a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \\ a_{41} & a_{42} & a_{43} \end{vmatrix}.$$

Each 3×3 determinant can be expanded using the 3×3 formulas above.

Useful properties

For any $n \times n$ matrix A :

- $\det(A) = 0$ if and only if A is singular (not invertible).
- Swapping two rows multiplies the determinant by -1 .
- Multiplying a row by a scalar k multiplies the determinant by k .
- Adding a multiple of one row to another row does not change the determinant.
- $\det(AB) = \det(A)\det(B)$ for any $n \times n$ matrices A, B .
- $\det(A^T) = \det(A)$.

Examples↓

Example 1: 2×2 determinant

Compute the determinant of

$$A = \begin{bmatrix} 2 & -3 \\ 4 & 1 \end{bmatrix}.$$

Using the formula $\det(A) = ad - bc$, we have

$$\det(A) = (2)(1) - (-3)(4)$$

$$\det(A) = 2 - (-12)$$

$$\det(A) = 2 + 12 = 14.$$

Example 2: 3×3 determinant (cofactor expansion)

Compute the determinant of

$$B = \begin{bmatrix} 1 & 2 & 3 \\ 0 & -1 & 4 \\ 2 & 1 & 0 \end{bmatrix}.$$

Expand along the first row:

$$\det(B) = 1 \begin{vmatrix} -1 & 4 \\ 1 & 0 \end{vmatrix} - 2 \begin{vmatrix} 0 & 4 \\ 2 & 0 \end{vmatrix} + 3 \begin{vmatrix} 0 & -1 \\ 2 & 1 \end{vmatrix}.$$

Now evaluate each 2×2 determinant:

$$\begin{vmatrix} -1 & 4 \\ 1 & 0 \end{vmatrix} = (-1)(0) - (4)(1) = -4,$$

$$\begin{vmatrix} 0 & 4 \\ 2 & 0 \end{vmatrix} = (0)(0) - (4)(2) = -8,$$

$$\begin{vmatrix} 0 & -1 \\ 2 & 1 \end{vmatrix} = (0)(1) - (-1)(2) = 2.$$

Substitute back:

$$\det(B) = 1(-4) - 2(-8) + 3(2)$$

$$\det(B) = -4 + 16 + 6$$

$$\det(B) = 18.$$

Example 3: 3×3 determinant (Sarrus' rule)

Compute the determinant of

$$C = \begin{bmatrix} 2 & 0 & 1 \\ -1 & 3 & 2 \\ 4 & -2 & 1 \end{bmatrix}.$$

Using Sarrus' rule, write the first two columns again:

$$\begin{array}{ccc|cc} 2 & 0 & 1 & 2 & 0 \\ -1 & 3 & 2 & -1 & 3 \\ 4 & -2 & 1 & 4 & -2 \end{array}$$

Sum of downward diagonals:

$$(2)(3)(1) + (0)(2)(4) + (1)(-1)(-2) = 6 + 0 + 2 = 8.$$

Sum of upward diagonals:

$$(1)(3)(4) + (2)(2)(2) + (0)(-1)(1) = 12 + 8 + 0 = 20.$$

Therefore,

$$\det(C) = 8 - 20 = -12.$$

Example 4: 4×4 determinant with zeros

Compute the determinant of

$$D = \begin{bmatrix} 1 & 0 & 2 & -1 \\ 3 & 1 & 0 & 2 \\ 0 & -2 & 1 & 0 \\ 2 & 0 & 1 & 1 \end{bmatrix}.$$

Expand along the first row (because it has a zero):

$$\det(D) = 1 \cdot C_{11} + 0 \cdot C_{12} + 2 \cdot C_{13} + (-1) \cdot C_{14}.$$

So

$$\det(D) = 1 \cdot \begin{vmatrix} 1 & 0 & 2 \\ -2 & 1 & 0 \\ 0 & 1 & 1 \end{vmatrix} + 2 \cdot \begin{vmatrix} 3 & 1 & 2 \\ 0 & -2 & 0 \\ 2 & 0 & 1 \end{vmatrix} - 1 \cdot \begin{vmatrix} 3 & 1 & 0 \\ 0 & -2 & 1 \\ 2 & 0 & 1 \end{vmatrix}.$$

Call these three 3×3 determinants D_1 , D_2 , and D_3 .

Compute D_1 :

$$D_1 = \begin{vmatrix} 1 & 0 & 2 \\ -2 & 1 & 0 \\ 0 & 1 & 1 \end{vmatrix}.$$

Expand along the first row:

$$D_1 = 1 \begin{vmatrix} 1 & 0 \\ 1 & 1 \end{vmatrix} - 0 \begin{vmatrix} -2 & 0 \\ 0 & 1 \end{vmatrix} + 2 \begin{vmatrix} -2 & 1 \\ 0 & 1 \end{vmatrix}.$$

Compute the 2×2 determinants:

$$\begin{vmatrix} 1 & 0 \\ 1 & 1 \end{vmatrix} = (1)(1) - (0)(1) = 1,$$

$$\begin{vmatrix} -2 & 1 \\ 0 & 1 \end{vmatrix} = (-2)(1) - (1)(0) = -2.$$

Thus

$$D_1 = 1(1) + 2(-2) = 1 - 4 = -3.$$

Compute D_2 :

$$D_2 = \begin{vmatrix} 3 & 1 & 2 \\ 0 & -2 & 0 \\ 2 & 0 & 1 \end{vmatrix}.$$

Expand along the second row (it has a zero):

$$D_2 = 0 \cdot \begin{vmatrix} 1 & 2 \\ 0 & 1 \end{vmatrix} - (-2) \cdot \begin{vmatrix} 3 & 2 \\ 2 & 1 \end{vmatrix} + 0 \cdot \begin{vmatrix} 3 & 1 \\ 2 & 0 \end{vmatrix}.$$

Only the middle term remains:

$$D_2 = 2 \begin{vmatrix} 3 & 2 \\ 2 & 1 \end{vmatrix}.$$

Compute the 2×2 determinant:

$$\begin{vmatrix} 3 & 2 \\ 2 & 1 \end{vmatrix} = (3)(1) - (2)(2) = 3 - 4 = -1.$$

So

$$D_2 = 2(-1) = -2.$$

Compute D_3 :

$$D_3 = \begin{vmatrix} 3 & 1 & 0 \\ 0 & -2 & 1 \\ 2 & 0 & 1 \end{vmatrix}.$$

Expand along the first row (because of the zero):

$$D_3 = 3 \begin{vmatrix} -2 & 1 \\ 0 & 1 \end{vmatrix} - 1 \begin{vmatrix} 0 & 1 \\ 2 & 1 \end{vmatrix} + 0 \cdot \begin{vmatrix} 0 & -2 \\ 2 & 0 \end{vmatrix}.$$

Compute the 2×2 determinants:

$$\begin{vmatrix} -2 & 1 \\ 0 & 1 \end{vmatrix} = (-2)(1) - (1)(0) = -2,$$

$$\begin{vmatrix} 0 & 1 \\ 2 & 1 \end{vmatrix} = (0)(1) - (1)(2) = -2.$$

Thus

$$D_3 = 3(-2) - 1(-2) = -6 + 2 = -4.$$

Combine to get $\det(D)$:

$$\det(D) = 1 \cdot D_1 + 2 \cdot D_2 - 1 \cdot D_3$$

$$\det(D) = (-3) + 2(-2) - (-4)$$

$$\det(D) = -3 - 4 + 4 = -3.$$

So

$$\det(D) = -3.$$